TABLA DE PARES DE TRANSFORMADA DE LAPLACE

	$f(t)$	$F(s)$
1	Impulso unitario $\delta(t)$	1
2	Escalón unitario $1(t)$	$\frac{1}{s}$
3	t	$\frac{1}{s^2}$
4	$\frac{t^{n-1}}{(n-1)!}$ (n=1,2,3,…)	$\frac{1}{s^n}$
5	t^n (n=1,2,3,…)	$\frac{n!}{s^{n+1}}$
6	e^{-at}	$\frac{1}{s+a}$
7	te^{-at}	$\frac{1}{(s+a)^2}$
8	$\frac{1}{(n-1)!}t^{n-1}e^{-at}$ (n=1,2,3,…)	$\frac{1}{(s+a)^n}$
9	$t^n e^{-at}$ (n=1,2,3,…)	$\frac{n!}{(s+a)^{n+1}}$
10	$\sin wt$	$\frac{w}{s^2+w^2}$
11	$\cos wt$	$\frac{s}{s^2+w^2}$
12	$\sinh wt$	$\frac{w}{s^2-w^2}$
13	$\cosh wt$	$\frac{s}{s^2-w^2}$
14	$\frac{1}{a}(1-e^{-at})$	$\frac{1}{s(s+a)}$
15	$\frac{1}{b-a}(e^{-at}-e^{-bt})$	$\frac{1}{(s+a)(s+b)}$
16	$\frac{1}{b-a}(be^{-bt}-ae^{-at})$	$\frac{s}{(s+a)(s+b)}$
17	$\frac{1}{ab}[1+\frac{1}{a-b}(be^{-at}-ae^{bt})]$	$\frac{1}{s(s+a)(s+b)}$
18	$\frac{1}{a^2}(1-e^{-at}-ate^{-at})$	$\frac{1}{s(s+a)^2}$
19	$\frac{1}{a^2}(at-1+e^{-at})$	$\frac{1}{s^2(s+a)}$
20	$e^{-at}\sin wt$	$\frac{w}{(s+a)^2+w^2}$

21	$e^{-at}\cos wt$	$\frac{s+a}{(s+a)^2+w^2}$
22	$\frac{w_n}{\sqrt{1-\zeta^2}}e^{-\zeta w_n t}\sin w_n\sqrt{1-\zeta^2}t$	$\frac{w_n^2}{s^2+2\zeta w_n s+w_n^2}$
23	$-\frac{1}{\sqrt{1-\zeta^2}}e^{-\zeta w_n t}\sin(w_n\sqrt{1-\zeta^2}t+\Phi)$ $\Phi=\tan^{-1}\frac{\sqrt{1-\zeta^2}}{\zeta}$	$\frac{s}{s^2+2\zeta w_n s+w_n^2}$
24	$1-\frac{1}{\sqrt{1-\zeta^2}}e^{-\zeta w_n t}\sin(w_n\sqrt{1-\zeta^2}t+\Phi)$ $\Phi=\tan^{-1}\frac{\sqrt{1-\zeta^2}}{\zeta}$	$\frac{w_n^2}{s^2+2\zeta w_n s+w_n^2}$
25	$1-\cos wt$	$\frac{w^2}{s(s^2+w^2)}$
26	$wt-\sin wt$	$\frac{w^3}{s^2(s^2+w^2)}$
27	$\sin wt-wt\cos wt$	$\frac{2w^3}{(s^2+w^2)^2}$
28	$\frac{1}{2w}t\sin wt$	$\frac{s}{(s^2+w^2)^2}$
29	$t\cos wt$	$\frac{s^2-w^2}{(s^2+w^2)^2}$
30	$\frac{1}{w_2^2-w_1^2}(\cos w_1 t-\cos w_2 t)\ \ (w_1^2\neq w_2^2)$	$\frac{s}{(s^2+w_1^2)(s^2+w_2^2)}$
31	$\frac{1}{2w}(\sin wt+wt\cos wt)$	$\frac{s^2}{(s^2+w^2)^2}$

TABLA DE PROPIEDADES DE LAS TRANSFORMADAS DE LAPLACE

1	$\mathcal{L}[AF(t)] = Af(s)$
2	$\mathcal{L}[F_1(t) \pm F_2(t)] = f_1(s) \pm f_2(s)$
3	$\mathcal{L}_{\pm}\left[\frac{d}{dt}F(t)\right] = sf(s) - F(0)$
4	$\mathcal{L}_{\pm}\left[\frac{d^2}{dt^2}F(t)\right] = s^2 f(s) - sF(0) - F'(0)$
5	$\mathcal{L}_{\pm}\left[\frac{d^n}{dt^n}F(t)\right] = s^n f(s) - \sum_{k=1}^{n} s^{n-k}\ F^{(k-1)}(0)$ donde $F^{(k-1)}(t) = \frac{d^{k-1}}{dt^{k-1}}F(t)$
6	$\mathcal{L}_{\pm}\left[\int F(t)dt\right] = \frac{f(s)}{s} + \frac{1}{s}[\int F(t)dt]_{t\neq 0\pm}$
7	$\mathcal{L}_{\pm}\left[\int \dots \int F(t)(dt)^n\right] = \frac{f(s)}{s^n} + \sum_{k=1}^{n} \frac{1}{s^{n-k+1}}[\int \dots \int F(t)(dt)^k]_{t=0\pm}$
8	$\mathcal{L}\left[\int_0^t F(t)dt\right] = \frac{f(s)}{s}$
9	$\int_0^{\infty} F(t)dt = \lim_{s\to 0} f(s)$ si $\int_0^{\infty} F(t)dt$ existe
10	$\mathcal{L}[e^{-at}F(t)] = f(s+a)$
11	$\mathcal{L}[F(t-a)l(t-a)] = e^{-\propto s}f(s)\ \ \propto \geq 0$
12	$\mathcal{L}[tF(t)] = -\frac{df(s)}{ds}$
13	$\mathcal{L}[t^2F(t)] = \frac{d^2}{ds^2}f(s)$
14	$\mathcal{L}[t^nF(t)] = (-1)^n\frac{d^n}{ds^n}f(s) \quad n = 1,2,3,\dots$
15	$\mathcal{L}\left[\frac{1}{t}F(t)\right] = \int_s^{\infty} f(s)ds$ si $\lim_{s\to 0}\frac{1}{t}F(t)\ existe$
16	$\mathcal{L}\left[F\left(\frac{t}{a}\right)\right] = af(as)$
17	$\mathcal{L}\left[\int_0^t F_1(t-\tau)F_2(\tau)d\tau\right] = f_1(s)f_2(s)$
18	$\mathcal{L}[F(t)G(t)] = \frac{1}{2\pi j}\int_{e-j\infty}^{e+j\infty} f(p)g(s-p)dp$

SERIE E INTEGRAL DE FOURIER – TRANSFORMADA DE FOURIER

	$f(t)$	$F(w)$
1	$h(t)$	$\frac{1}{iw}$
2	$h(t)e^{-at}\ ;\ a>0$	$\frac{1}{a+iw}$
3	$e^{-a\lvert t\rvert}\ ;\ a>0$	$\frac{2a}{a^2+\ w^2}$
4	$e^{-at^2}\ ;\ a>0$	$\sqrt{\frac{\pi}{a}}\ e^{-w^2/4a}$
5	$h(t+a)-h(t-a)$	$F(w)=\frac{2}{w}\sin(aw)$
6	$\delta(t)$	1

Propiedades de la Transformada de Fourier

1	$\mathcal{F}\{c_1 f(t) + c_2 g(t)\} = c_1 \mathcal{F}\{f(t)\} + c_2 \mathcal{F}\{g(t)\} = c_1 F(w) + c_2 G(w)$
2	$\mathcal{F}\{f^{(n)}(t)\} = (iw)^n F(w)$
3	$\mathcal{F}\{e^{iw_0 t} f(t)\} = F(w - w_0)$
4	$\mathcal{F}\{f(t - t_0)\} = e^{-t_0 iw} F(w)$
5	$\mathcal{F}\{t^n f(t)\} = i^n F^{(n)}(w) \quad n \in \mathbb{N}$
6	$f(t) * g(t) = \int_{\infty-}^{\infty+} f(\tau) g(t - \tau) d\tau$

TABLAS DE DERIVADAS

	FUNCIÓN SIMPLE – DERIVADAS INMEDIATAS		FUNCIÓN COMPUESTA – REGLA DE LA CADENA	
	FUNCIÓN	**DERIVADA**	**FUNCIÓN**	**DERIVADA**
	$f(x)$	$f'(x)$	$f(x)=f(u(x))$ $u=u(x)$	$f'(x)=f'(u)\cdot u'(x)$
1	$f(x)=c$	$f'(x)=0$		
2	$f(x)=x$	$f'(x)=1$	$f(x)=u$	$f'(x)=u'$
3	$f(x)=c\cdot x$	$f'(x)=c$	$f(x)=c\cdot u$	$f'(x)=c\cdot u'$
4			$f(x)=u\pm v\pm w$	$f'(x)=u'\pm v'\pm w'$
5			$f(x)=u\cdot v$	$f'(x)=u'v+uv'$
6			$f(x)=\dfrac{u}{v}$	$f'(x)=\dfrac{u'v-uv'}{v^2}$
7			$f(x)=u^v$	$f'(x)=u^v\cdot\left(v'\cdot\ln u+v\cdot\dfrac{u'}{u}\right)$
8	$f(x)=x^n$	$f'(x)=n\cdot x^{n-1}$	$f(x)=u^n$	$f'(x)=n\cdot u^{n-1}\cdot u'$
9	$f(x)=\sqrt[n]{x}$	$f'(x)=\dfrac{1}{n\cdot\sqrt[n]{x^{n-1}}}$	$f(x)=\sqrt[n]{u}$	$f'(x)=\dfrac{u'}{n\cdot\sqrt[n]{u^{n-1}}}$
10	$f(x)=\log_a x$	$f'(x)=\dfrac{1}{x}\log_a e=\dfrac{1}{x\ln a}$	$f(x)=\log_a u$	$f'(x)=\dfrac{u'}{u}\log_a e=\dfrac{u'}{u\ln a}$
11	$f(x)=\ln x$	$f'(x)=\dfrac{1}{x}$	$f(u)=\ln u$	$f'(x)=\dfrac{u'}{u}$
12	$f(x)=e^x$	$f'(x)=e^x$	$f(x)=e^u$	$f'(x)=e^u\cdot u'$
13	$f(x)=a^x$	$f'(x)=a^x\ln a$	$f(x)=a^u$	$f'(x)=a^u\cdot u'\cdot\ln a$
15	$f(x)=\operatorname{sen} x$	$f'(x)=\cos x$	$f(x)=\operatorname{sen} u$	$f'(x)=u'\cos u$
16	$f(x)=\cos x$	$f'(x)=-\operatorname{sen} x$	$f(x)=\cos u$	$f'(x)=-u'\operatorname{sen} u$
17	$f(x)=\operatorname{tg} x$	$f'(x)=\sec^2 x=\dfrac{1}{\cos^2 x}=1+\operatorname{tg}^2 x$	$f(x)=\operatorname{tg} u$	$f'(x)=u'\sec^2 u=\dfrac{u'}{\cos^2 u}=(1+\operatorname{tg}^2 u)u'$
18	$f(x)=\operatorname{cotg} x$	$f'(x)=-\operatorname{cosec}^2 x=\dfrac{-1}{\operatorname{sen}^2 x}=-(1+\operatorname{cotg}^2 x)$	$f(x)=\operatorname{cotg} u$	$f'(x)=-u'\cdot\operatorname{cosec}^2 u=\dfrac{-u'}{\operatorname{sen}^2 u}=-u'\cdot(1+\operatorname{cotg}^2 u)$
19	$f(x)=\sec x$	$f'(x)=\sec x\cdot\operatorname{tg} x$	$f(x)=\sec u$	$f'(x)=u'\cdot\sec u\cdot\operatorname{tg} u$
20	$f(x)=\operatorname{cosec} x$	$f'(x)=-\operatorname{cosec} x\cdot\operatorname{cotg} x$	$f(x)=\operatorname{cosec} u$	$f'(x)=-u'\cdot\operatorname{cosec} u\cdot\operatorname{cotg} u$
21	$f(x)=\operatorname{arcosen} x$	$f'(x)=\dfrac{1}{\sqrt{1-x^2}}$	$f(x)=\operatorname{arcosen} u$	$f'(x)=\dfrac{u'}{\sqrt{1-u^2}}$
22	$f(x)=\operatorname{arcocos} x$	$f'(x)=\dfrac{-1}{\sqrt{1-x^2}}$	$f(x)=\operatorname{arcocos} u$	$f'(x)=\dfrac{-u'}{\sqrt{1-u^2}}$

23	$f(x)=\operatorname{arcotg} x$	$f'(x)=\dfrac{1}{1+x^2}$	$f(x)=\operatorname{arcotg} u$	$f'(x)=\dfrac{u'}{1+u^2}$
24	$f(x)=\operatorname{arcocotg} x$	$f'(x)=\dfrac{-1}{1+x^2}$	$f(x)=\operatorname{arcocotg} u$	$f'(x)=\dfrac{-u'}{1+u^2}$
25	$f(x)=\operatorname{arcosec} x$	$f'(x)=\dfrac{1}{x\cdot\sqrt{x^2-1}}$	$f(x)=\operatorname{arcosec} u$	$f'(x)=\dfrac{u'}{u\cdot\sqrt{u^2-1}}$
26	$f(x)=\operatorname{arcocosec} x$	$f'(x)=\dfrac{-1}{x\cdot\sqrt{x^2-1}}$	$f(x)=\operatorname{arcocosec} u$	$f'(x)=\dfrac{-u'}{u\cdot\sqrt{u^2-1}}$
27	$f(x)=\lvert x\rvert$	$f'(x)=\dfrac{x}{\lvert x\rvert}\quad \forall\, x\neq 0$	$f(x)=\lvert u\rvert$	$f'(x)=\dfrac{u}{\lvert u\rvert}\quad \forall\, u\neq 0$

TABLAS DE INTEGRALES

	INTEGRALES INMEDIATAS		INTEGRALES POR SUSTITUCIÓN	
	FUNCIÓN SIMPLE		FUNCIÓN COMPUESTA	
			$u = f(x) \quad \frac{du}{dx} = u' = f'(x) \to du = f'(x)dx$	
1	$\int (f_1(x) \pm f_2(x) \pm \ldots \pm f_n(x))dx = \int f_1(x)dx \pm \int f_2(x)dx \pm \ldots \pm \int f_n(x)dx$			
2	$\int udv = u \cdot v - \int vdu$			
3	$\int dx$	$x + C$	$\int du$	$u + C$
4	$\int c\,dx$	$c \cdot x + C$	$\int c\,du$	$c \cdot u + C$
5	$\int x^n dx$	$\frac{x^{n+1}}{n+1} + C \quad \forall\, n \neq -1$	$\int u^n du$	$\frac{u^{n+1}}{n+1} + C \quad \forall\, n \neq -1$
6	$\int \frac{1}{n \cdot \sqrt[n]{x^{n-1}}} dx$	$\sqrt[n]{x} + C$	$\int \frac{1}{n \cdot \sqrt[n]{u^{n-1}}} du$	$\sqrt[n]{u} + C$
7	$\int \frac{1}{x} dx$	$\ln x + C$	$\int \frac{1}{u} du$	$\ln\lvert u\rvert + C$
8	$\int e^x dx$	$e^x + C$	$\int e^u du$	$e^u + C$
9	$\int a^x dx$	$\frac{a^x}{\ln a} + C$	$\int a^u du$	$\frac{a^u}{\ln a} + C$
10	$\int \operatorname{sen} x\,dx$	$-\cos x + C$	$\int \operatorname{sen} u\,du$	$-\cos u + C$
11	$\int \cos x\,dx$	$\operatorname{sen} x + C$	$\int \cos u\,du$	$\operatorname{sen} u + C$
12	$\int \sec^2 x\,dx$ $\int \frac{1}{\cos^2 x} dx$ $\int (1 + \operatorname{tg}^2 x)dx$	$\operatorname{tg} x + C$	$\int \sec^2 u\,du$ $\int \frac{1}{\cos^2 u} du$	$\operatorname{tg} u + C$
13	$\int \operatorname{cosec}^2 x\,dx$ $\int \frac{1}{\operatorname{sen}^2 x} dx$ $\int (1 + \operatorname{cotg}^2 x)dx$	$-\operatorname{cotg} x + C$	$\int \operatorname{cosec}^2 u\,du$ $\int \frac{1}{\operatorname{sen}^2 u} du$	$-\operatorname{cotg} u + C$
14	$\int \sec x\,dx$	$\ln\lvert\sec x + \operatorname{tg} x\rvert + C$	$\int \sec u\,du$	$\ln\lvert\sec u + \operatorname{tg} u\rvert + C$
15	$\int \operatorname{cosec} x\,dx$	$\ln\lvert\operatorname{cosec} x - \operatorname{cotg} x\rvert + C$	$\int \operatorname{cosec} u\,du$	$\ln\lvert\operatorname{cosec} u - \operatorname{cotg} u\rvert + C$
16	$\int \sec x \cdot \operatorname{tg} x\,dx$	$\sec x + C$	$\int \sec u \cdot \operatorname{tg} u\,du$	$\sec u + C$
17	$\int \operatorname{cosec} x \cdot \operatorname{cotg} x\,dx$	$-\operatorname{cosec} x + C$	$\int \operatorname{cosec} u \cdot \operatorname{cotg} u\,du$	$-\operatorname{cosec} u + C$

18	$\int \frac{1}{\sqrt{1-x^2}}dx$	$\operatorname{arcosen} x + C$	$\int \frac{1}{\sqrt{1-u^2}}du$ $\int \frac{1}{\sqrt{a^2-u^2}}du$	$\operatorname{arcosen} u + C$ $\operatorname{arcosen}\frac{u}{a} + C$
19	$\int \frac{1}{1+x^2}dx$	$\operatorname{arcotg} x + C$	$\int \frac{1}{1+u^2}du$ $\int \frac{1}{a^2+u^2}du$	$\operatorname{arcotg} u + C$ $\frac{1}{a}\operatorname{arcotg}\frac{u}{a} + C$
20	$\int \frac{1}{x\cdot\sqrt{x^2-1}}dx$	$\operatorname{arcosec} x + C$	$\int \frac{1}{u\cdot\sqrt{u^2-1}}du$ $\int \frac{1}{u\cdot\sqrt{u^2-a^2}}du$	$\operatorname{arcosec} u + C$ $\frac{1}{a}\operatorname{arcosec}\frac{u}{a} + C$

INTEGRALES ADICIONALES

Formas que involucran $\sqrt{a^2+u^2}\,du,\ a>0$

$$\int \sqrt{a^2+u^2}\,du = \frac{u}{2}\sqrt{a^2+u^2} + \frac{a^2}{2}\ln\left(u+\sqrt{a^2+u^2}\right) + C$$

$$\int u^2\sqrt{a^2+u^2}\,du = \frac{u}{8}(a^2+2u^2)\sqrt{a^2+u^2} - \frac{a^4}{8}\ln\left(u+\sqrt{a^2+u^2}\right) + C$$

$$\int \frac{\sqrt{a^2+u^2}}{u}\,du = \sqrt{a^2+u^2} - a\ln\left|\frac{a+\sqrt{a^2+u^2}}{u}\right| + C$$

$$\int \frac{\sqrt{a^2+u^2}}{u^2}\,du = -\frac{\sqrt{a^2+u^2}}{u} + \ln\left(u+\sqrt{a^2+u^2}\right) + C$$

$$\int \frac{du}{\sqrt{a^2+u^2}} = \ln\left(u+\sqrt{a^2+u^2}\right) + C$$

$$\int \frac{u^2\,du}{\sqrt{a^2+u^2}} = \frac{u}{2}\sqrt{a^2+u^2} - \frac{a^2}{2}\ln\left(u+\sqrt{a^2+u^2}\right) + C$$

$$\int \frac{du}{u\sqrt{a^2+u^2}} = -\frac{1}{a}\ln\left|\frac{\sqrt{a^2+u^2}+a}{u}\right| + C$$

$$\int \frac{du}{u^2\sqrt{a^2+u^2}} = -\frac{\sqrt{a^2+u^2}}{a^2u} + C$$

$$\int \frac{du}{(a^2+u^2)^{3/2}} = \frac{u}{a^2\sqrt{a^2+u^2}} + C$$

Formas que involucran $\sqrt{a^2-u^2},\ a>0$

$$\int \sqrt{a^2-u^2}\,du = \frac{u}{2}\sqrt{a^2-u^2} + \frac{a^2}{2}\operatorname{sen}^{-1}\frac{u}{a} + C$$

$$\int u^2\sqrt{a^2-u^2}\,du = \frac{u}{8}(2u^2-a^2)\sqrt{a^2-u^2} + \frac{a^4}{8}\operatorname{sen}^{-1}\frac{u}{a} + C$$

$$\int \frac{\sqrt{a^2-u^2}}{u}\,du = \sqrt{a^2-u^2} - a\ln\left|\frac{a+\sqrt{a^2-u^2}}{u}\right| + C$$

$$\int \frac{\sqrt{a^2-u^2}}{u^2}\,du = -\frac{1}{u}\sqrt{a^2-u^2} - \operatorname{sen}^{-1}\frac{u}{a} + C$$

$$\int \frac{u^2\,du}{\sqrt{a^2-u^2}} = -\frac{u}{2}\sqrt{a^2-u^2} + \frac{a^2}{2}\operatorname{sen}^{-1}\frac{u}{a} + C$$

$$\int \frac{du}{u\sqrt{a^2-u^2}} = -\frac{1}{a}\ln\left|\frac{a+\sqrt{a^2-u^2}}{u}\right| + C$$

$$\int \frac{du}{u^2\sqrt{a^2-u^2}} = -\frac{1}{a^2u}\sqrt{a^2-u^2} + C$$

$$\int (a^2-u^2)^{3/2}\,du = -\frac{u}{8}(2u^2-5a^2)\sqrt{a^2-u^2} + \frac{3a^4}{8}\operatorname{sen}^{-1}\frac{u}{a} + C$$

$$\int \frac{du}{(a^2-u^2)^{3/2}} = \frac{u}{a^2\sqrt{a^2-u^2}} + C$$

Formas que involucran $\sqrt{u^2-a^2}$, $a>0$

$$\int \sqrt{u^2-a^2}\,du = \frac{u}{2}\sqrt{u^2-a^2} - \frac{a^2}{2}\ln\left|u+\sqrt{u^2-a^2}\right| + C$$

$$\int u^2\sqrt{u^2-a^2}\,du = \frac{u}{8}(2u^2-a^2)\sqrt{u^2-a^2} - \frac{a^4}{8}\ln\left|u+\sqrt{u^2-a^2}\right| + C$$

$$\int \frac{\sqrt{u^2-a^2}}{u}\,du = \sqrt{u^2-a^2} - a\cos^{-1}\frac{a}{|u|} + C$$

$$\int \frac{\sqrt{u^2-a^2}}{u^2}\,du = -\frac{\sqrt{u^2-a^2}}{u} + \ln\left|u+\sqrt{u^2-a^2}\right| + C$$

$$\int \frac{du}{\sqrt{u^2-a^2}} = \ln\left|u+\sqrt{u^2-a^2}\right| + C$$

$$\int \frac{u^2\,du}{\sqrt{u^2-a^2}} = \frac{u}{2}\sqrt{u^2-a^2} + \frac{a^2}{2}\ln\left|u+\sqrt{u^2-a^2}\right| + C$$

$$\int \frac{du}{u^2\sqrt{u^2-a^2}} = \frac{\sqrt{u^2-a^2}}{a^2u} + C$$

$$\int \frac{du}{(u^2-a^2)^{3/2}} = -\frac{u}{a^2\sqrt{u^2-a^2}} + C$$

Formas que involucran $a+bu$

$$\int u\sqrt{a+bu}\,du = \frac{2}{15b^2}(3bu-2a)(a+bu)^{3/2} + C$$

$$\int \frac{u\,du}{\sqrt{a+bu}} = \frac{2}{3b^2}(bu-2a)\sqrt{a+bu} + C$$

$$\int \frac{u^2\,du}{\sqrt{a+bu}} = \frac{2}{15b^3}(8a^2 + 3b^2u^2 - 4abu)\sqrt{a+bu} + C$$

$$\int \frac{du}{u\sqrt{a+bu}} = \frac{1}{\sqrt{a}} \ln \left| \frac{\sqrt{a+bu} - \sqrt{a}}{\sqrt{a+bu} + \sqrt{a}} \right| + C, \quad \text{si } a > 0$$

$$= \frac{2}{\sqrt{-a}} \tan^{-1} \sqrt{\frac{a+bu}{-a}} + C, \quad \text{si } a < 0$$

$$\int \frac{\sqrt{a+bu}}{u}\,du = 2\sqrt{a+bu} + a\int \frac{du}{u\sqrt{a+bu}}$$

$$\int \frac{\sqrt{a+bu}}{u^2}\,du = -\frac{\sqrt{a+bu}}{u} + \frac{b}{2}\int \frac{du}{u\sqrt{a+bu}}$$

$$\int u^n\sqrt{a+bu}\,du = \frac{2}{b(2n+3)}\left[u^n(a+bu)^{3/2} - na\int u^{n-1}\sqrt{a+bu}\,du\right]$$

$$\int \frac{u^n\,du}{\sqrt{a+bu}} = \frac{2u^n\sqrt{a+bu}}{b(2n+1)} - \frac{2na}{b(2n+1)}\int \frac{u^{n-1}\,du}{\sqrt{a+bu}}$$

$$\int \frac{du}{u^n\sqrt{a+bu}} = -\frac{\sqrt{a+bu}}{a(n-1)u^{n-1}} - \frac{b(2n-3)}{2a(n-1)}\int \frac{du}{u^{n-1}\sqrt{a+bu}}$$

Formas que involucran $a+bu+c^2, b^2 \neq 4ac$

$$\int \frac{1}{a+bu+cu^2}\,du = \begin{cases} \dfrac{2}{\sqrt{4ac-b^2}} \arctan \dfrac{2cu+b}{\sqrt{4ac-b^2}} + C, & b^2 < 4ac \\ \dfrac{1}{\sqrt{b^2-4ac}} \ln \left| \dfrac{2cu+b-\sqrt{b^2-4ac}}{2cu+b+\sqrt{b^2-4ac}} \right| + C, & b^2 > 4ac \end{cases}$$

$$\int \frac{u}{a+bu+cu^2}\,du = \frac{1}{2c}\left(\ln|a+bu+cu^2| - b\int \frac{1}{a+bu+cu^2}\,du\right)$$

Límites y Trigonometría

Tabla de referencia: propiedades de los límites, límites notables e identidades trigonométricas

1. Propiedades de los límites

Válidas cuando lim x→a f(x) = A y lim x→a g(x) = B existen y son finitos. Las fórmulas están construidas como ecuaciones nativas de Word (editables con el editor de ecuaciones).

Propiedad	Enunciado (dado lim x→a f(x) = A , lim x→a g(x) = B)
Límite de una constante	$lim_{x\to a}\, k = k$
Límite de x	$lim_{x\to a}\, x = a$
Múltiplo constante	$lim_{x\to a}\,[k \cdot f(x)] = k \cdot A$
Suma	$lim_{x\to a}\,[f(x) + g(x)] = A + B$
Diferencia	$lim_{x\to a}\,[f(x) - g(x)] = A - B$
Producto	$lim_{x\to a}\,[f(x) \cdot g(x)] = A \cdot B$
Cociente	$lim_{x\to a}\,\frac{[f(x)]}{[g(x)]} = \frac{A}{B}$ (B ≠ 0)
Potencia	$lim_{x\to a}\,[f(x)]^n = A^n$
Raíz	$lim_{x\to a}\,\sqrt[n]{f(x)} = \sqrt[n]{A}$ (A ≥ 0 si n es par)
Función polinómica	$lim_{x\to a}\, p(x) = p(a)$ (sustitución directa)
Función racional	$lim_{x\to a}\,\frac{p(x)}{q(x)} = \frac{p(a)}{q(a)}$ (q(a) ≠ 0)
Unicidad del límite	$Si\ el\ límite\ existe, es\ único$
Límites laterales	$lim_{x\to a}\, f(x) = L \iff lim_{x\to a^-}\, f(x) = lim_{x\to a^+}\, f(x) = L$
Función compuesta	$Si\ lim_{x\to a}\, g(x) = B\ \ y\ \ f\ es\ continua\ en\ B,$ $entonces\ \ lim_{x\to a}\, f(g(x)) = f(B)$
Teorema del emparedado (Sandwich)	$Si\ g(x) \le f(x) \le h(x)\ cerca\ de\ a,\quad y$ $lim_{x\to a}\, g(x) = lim_{x\to a}\, h(x) = L,\ \ entonces\ \ lim_{x\to a}\, f(x) = L$

2. Límites notables

Límite notable	Valor
$lim_{x\to 0}\,\frac{sen(x)}{x}$	1
$lim_{x\to 0}\,\frac{1 - cos\,x}{x}$	0
$lim_{x\to 0}\,\frac{1 - cos\,x}{x^2}$	$\frac{1}{2}$

Límite notable	Valor
$lim_{x\to 0}\ \frac{tan(x)}{x}$	1
$lim_{x\to 0}\ \frac{arcsen(x)}{x}$	1
$lim_{x\to 0}\ \frac{arctan(x)}{x}$	1
$lim_{x\to \infty}\ \left(1+\frac{1}{x}\right)^{x}$	e
$lim_{x\to 0}\ (1+x)^{\frac{1}{x}}$	e
$lim_{x\to \infty}\ \left(1+\frac{\alpha}{x}\right)^{x}$	e^{α}
$lim_{x\to 0}\ \frac{e^{x}-1}{x}$	1
$lim_{x\to 0}\ \frac{a^{x}-1}{x}$ (a > 0)	$ln(a)$
$lim_{x\to 0}\ \frac{ln(1+x)}{x}$	1
$lim_{x\to 0}\ \frac{log_{a}(1+x)}{x}$	$\frac{1}{ln(a)}$
$lim_{x\to +\infty}\ x^{n}$ (n > 0)	$+\infty$
$lim_{x\to -\infty}\ x^{n}$ (n par)	$+\infty$
$lim_{x\to -\infty}\ x^{n}$ (n impar)	$-\infty$
$lim_{x\to 0^{+}}\ \frac{1}{x^{n}}$	$+\infty$
$lim_{x\to +\infty}\ a^{x}$ (a > 1)	$+\infty$
$lim_{x\to -\infty}\ a^{x}$ (a > 1)	0
$lim_{x\to +\infty}\ a^{x}$ (0 < a < 1)	0
$lim_{x\to +\infty}\ ln(x)$	$+\infty$
$lim_{x\to 0^{+}}\ ln(x)$	$-\infty$
$lim_{x\to +\infty}\ \frac{ln(x)}{x}$	0
$lim_{x\to +\infty}\ \frac{x^{n}}{e^{x}}$	0
$lim_{x\to \infty}\ \frac{P(x)}{Q(x)}$ (grado P = grado Q)	$cociente\ de\ coef.principales$
$lim_{x\to \infty}\ \frac{P(x)}{Q(x)}$ (grado P < grado Q)	0
$lim_{x\to \infty}\ \frac{P(x)}{Q(x)}$ (grado P > grado Q)	$\pm\infty$

3. Identidades trigonométricas

3.1 Identidades pitagóricas

$sen^2\theta + cos^2\theta = 1$
$1 + tan^2\theta = sec^2\theta$
$1 + cot^2\theta = csc^2\theta$

3.2 Identidades recíprocas

$sen\,\theta = \frac{1}{csc\,\theta}$	$csc\,\theta = \frac{1}{sen\,\theta}$
$cos\,\theta = \frac{1}{sec\,\theta}$	$sec\,\theta = \frac{1}{cos\,\theta}$
$tan\,\theta = \frac{1}{cot\,\theta}$	$cot\,\theta = \frac{1}{tan\,\theta}$

3.3 Identidades de cociente

$tan\,\theta = \frac{sen\,\theta}{cos\,\theta}$
$cot\,\theta = \frac{cos\,\theta}{sen\,\theta}$

3.4 Identidades pares e impares

$sen(-\theta) = -sen\,\theta$	$cos(-\theta) = cos\,\theta$
$tan(-\theta) = -tan\,\theta$	$sec(-\theta) = sec\,\theta$
$csc(-\theta) = -csc\,\theta$	$cot(-\theta) = -cot\,\theta$

3.5 Identidades de cofunción

$sen\left(\frac{\pi}{2} - \theta\right) = cos\,\theta$	$cos\left(\frac{\pi}{2} - \theta\right) = sen\,\theta$
$tan\left(\frac{\pi}{2} - \theta\right) = cot\,\theta$	$cot\left(\frac{\pi}{2} - \theta\right) = tan\,\theta$
$sec\left(\frac{\pi}{2} - \theta\right) = csc\,\theta$	$csc\left(\frac{\pi}{2} - \theta\right) = sec\,\theta$

3.6 Identidades de suma y diferencia de ángulos

$sen(\alpha \pm \beta) = sen\,\alpha\,cos\,\beta \pm cos\,\alpha\,sen\,\beta$
$cos(\alpha \pm \beta) = cos\,\alpha\,cos\,\beta \mp sen\,\alpha\,sen\,\beta$

$$tan(\alpha \pm \beta) = \frac{tan\,\alpha \pm tan\,\beta}{1 \mp tan\,\alpha\,tan\,\beta}$$

3.7 Identidades de ángulo doble

$$sen(2\theta) = 2\,sen\,\theta\,cos\,\theta$$

$$cos(2\theta) = cos^2\theta - sen^2\theta = 2cos^2\theta - 1 = 1 - 2sen^2\theta$$

$$tan(2\theta) = \frac{2\,tan\,\theta}{1 - tan^2\theta}$$

3.8 Identidades de ángulo mitad

$$sen\left(\frac{\theta}{2}\right) = \pm\sqrt{\frac{1 - cos\,\theta}{2}}$$

$$cos\left(\frac{\theta}{2}\right) = \pm\sqrt{\frac{1 + cos\,\theta}{2}}$$

$$tan\left(\frac{\theta}{2}\right) = \pm\sqrt{\frac{1 - cos\,\theta}{1 + cos\,\theta}} = \frac{sen\,\theta}{1 + cos\,\theta} = \frac{1 - cos\,\theta}{sen\,\theta}$$

3.9 Producto a suma

$$sen\,\alpha\,cos\,\beta = \frac{1}{2}[sen(\alpha+\beta) + sen(\alpha-\beta)]$$

$$cos\,\alpha\,sen\,\beta = \frac{1}{2}[sen(\alpha+\beta) - sen(\alpha-\beta)]$$

$$cos\,\alpha\,cos\,\beta = \frac{1}{2}[cos(\alpha-\beta) + cos(\alpha+\beta)]$$

$$sen\,\alpha\,sen\,\beta = \frac{1}{2}[cos(\alpha-\beta) - cos(\alpha+\beta)]$$

3.10 Suma a producto

$$sen\,\alpha + sen\,\beta = 2\,sen\left(\frac{\alpha+\beta}{2}\right)cos\left(\frac{\alpha-\beta}{2}\right)$$

$$sen\,\alpha - sen\,\beta = 2\,cos\left(\frac{\alpha+\beta}{2}\right)sen\left(\frac{\alpha-\beta}{2}\right)$$

$$cos\,\alpha + cos\,\beta = 2\,cos\left(\frac{\alpha+\beta}{2}\right)cos\left(\frac{\alpha-\beta}{2}\right)$$

$$cos\,\alpha - cos\,\beta = -2\,sen\left(\frac{\alpha+\beta}{2}\right)sen\left(\frac{\alpha-\beta}{2}\right)$$

Fuentes

- OpenStax, Precálculo 2ed — 12.2 Hallar los límites: propiedades de los límites — https://openstax.org/books/prec%C3%A1lculo-2ed/pages/12-2-hallar-los-limites-propiedades-de-los-limites
- OpenStax, Cálculo Volumen 1 (vía LibreTexts en español) — Capítulo 2: Límites — https://espanol.libretexts.org/Bookshelves/Matematicas/Libro%3A_Calculo_(OpenStax)/02%3A_L%C3%ADmites
- LibreTexts en Español — 1.6: Propiedades de Límite — https://espanol.libretexts.org/Bookshelves/Educacion_Basica/Calculo/01%3A_L%C3%ADmite/1.06%3A_Propiedades_de_L%C3%ADmite
- Universidad Politécnica de Madrid (ETSISI) — Entrenamiento Matemático: Límites (material de apoyo, límites notables) — https://giematic.etsisi.upm.es/images/pdf/EntMat/EM-Sesion04-LIMITES-Apoyo.pdf
- OpenStax, Precálculo 2ed — 7.1 Resolver ecuaciones trigonométricas con identidades (pitagóricas, recíprocas, cociente, pares-impares) — https://openstax.org/books/prec%C3%A1lculo-2ed/pages/7-1-resolver-ecuaciones-trigonometricas-con-identidades
- LibreTexts en Español — 3.1.1: Identidades trigonométricas fundamentales — https://espanol.libretexts.org/Educacion_Basica/Trigonometria/03:_Identidades_trigonom%C3%A9tricas/3.01:_Identidades_trig/3.1.01:_Identidades_trigonom%C3%A9tricas_fundamentales
- LibreTexts en Español (Beveridge, Álgebra universitaria y trigonometría) — 10.1 Identidades recíprocas y pitagóreas / 10.2 Identidades de doble ángulo — https://espanol.libretexts.org/Bookshelves/Matematicas/Algebra/Libro%3A_Algebra_universitaria_y_trigonometria_(Beveridge)/10%3A_Identidades_trigonom%C3%A9tricas_y_ecuaciones/10.02%3A_Identidades_de_doble_%C3%A1ngulo
- Mathematics LibreTexts — Fundamental Trigonometric Identities (incluye producto-suma y suma-producto) — https://math.libretexts.org/Reference/Fundamental_Trigonometric_Identities

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